Gyro Logic: A Stability-Based Framework for Representation Under Intrinsic Deviation

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Abstract

Gyro Logic is a theoretical framework that models reality not as a direct reflection of an underlying structure, but as a reconfigured representation generated through observation. In this framework, observation is formalized as a Slice operation that transforms a latent Structure into a representation. Crucially, every Slice inherently introduces deviation (Δ), which is not treated as noise or error, but as an intrinsic component of representation.

Stability is defined as a functional of deviation, measuring the extent to which representations remain consistent across multiple observational transformations. Meaning, identity, and reference are then derived as stable structures that persist under controlled deviation. When deviation exceeds a critical threshold, the current Slice collapses, leading to a discontinuous transition (Jump) into a new representational regime. Furthermore, regions where deviation becomes undefined or unbounded are characterized as Void, representing the limit of evaluability.

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1. Introduction

Gyro Logic proposes that reality is not directly observed, but constructed through observation.

The framework is built on three core elements:

- Structure S
- Slice O
- Stability

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2. Core Architecture

$$X = O(S)$$

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3. Slice-Dependent Representation

$$X_1 = O_1(S), \quad X_2 = O_2(S)$$

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4. Stability as a Function of Deviation

Let the structure space be $\mathcal{S}$ and the representation space be $\mathcal{X}$.

$$O : \mathcal{S} \to \mathcal{X}$$

Define a metric:

$$d : \mathcal{X} \times \mathcal{X} \to \mathbb{R}_{\geq 0}$$

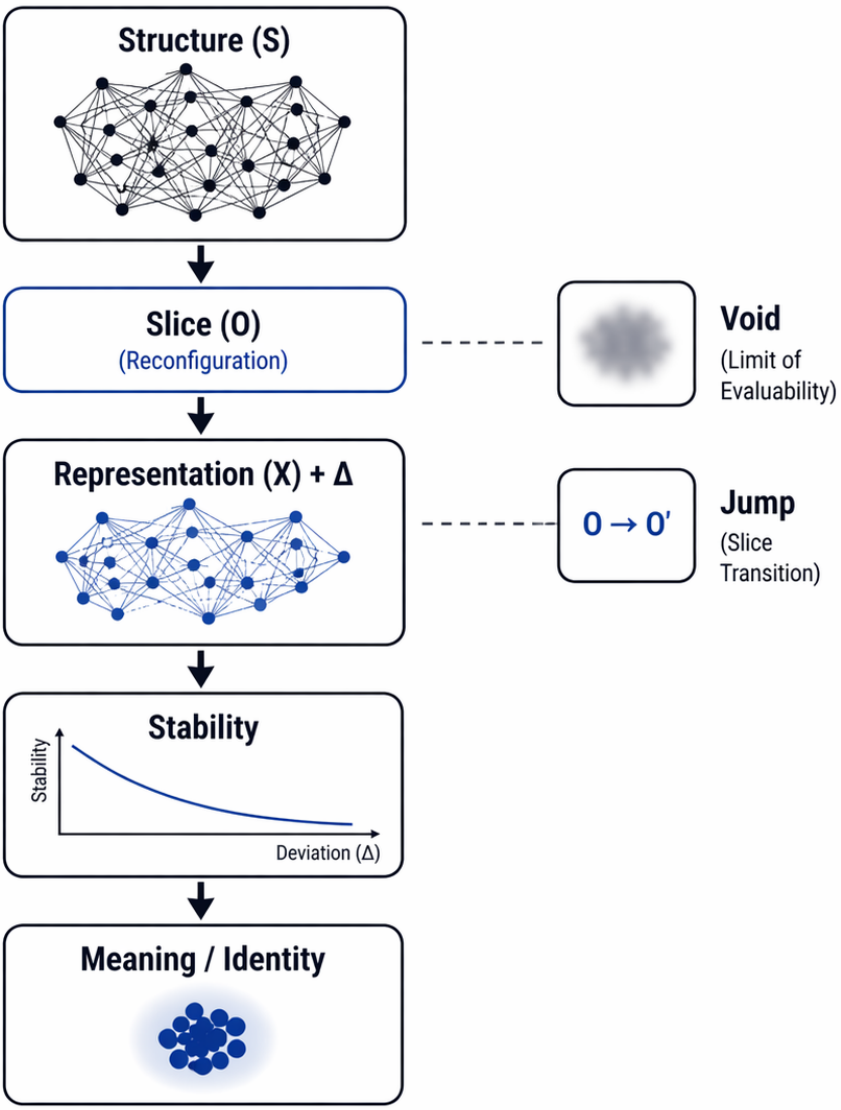

Figure 1. Core Architecture of Gyro Logic

Structure is reconfigured through Slice into a representation that inherently contains deviation (Δ). Stability evaluates whether the deviation remains within a tolerable range, determining the emergence of Meaning and Identity. Void represents the limit of evaluability, while Jump denotes a transition between slices.

Figure 1: Figure 1. Core Architecture of Gyro Logic

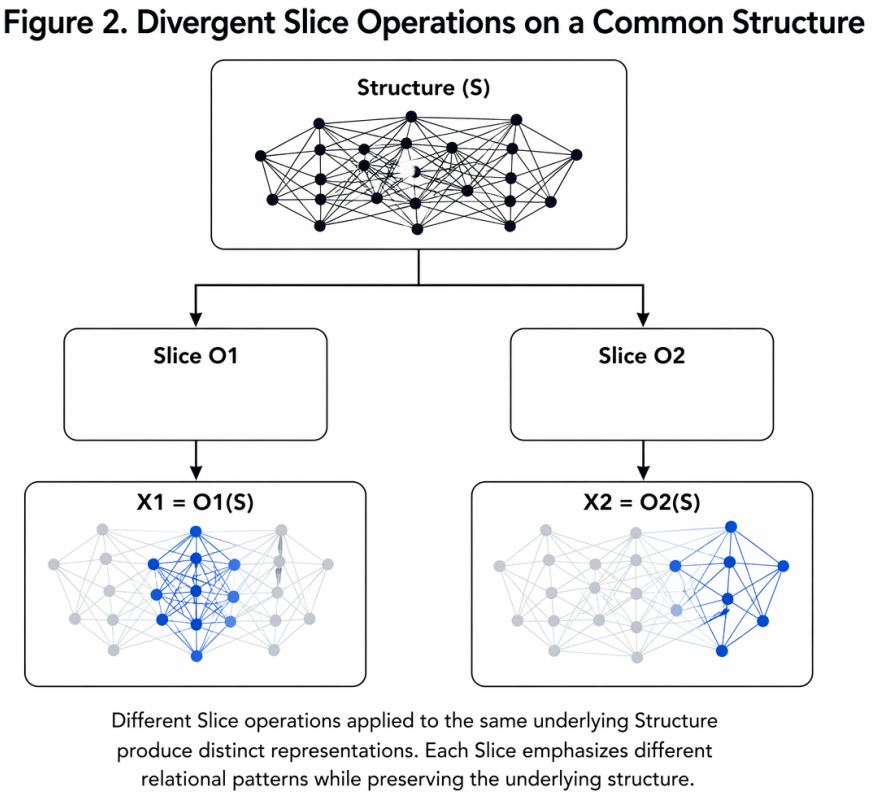

Figure 2: Figure 2. Divergent Slice Operations on a Common Structure

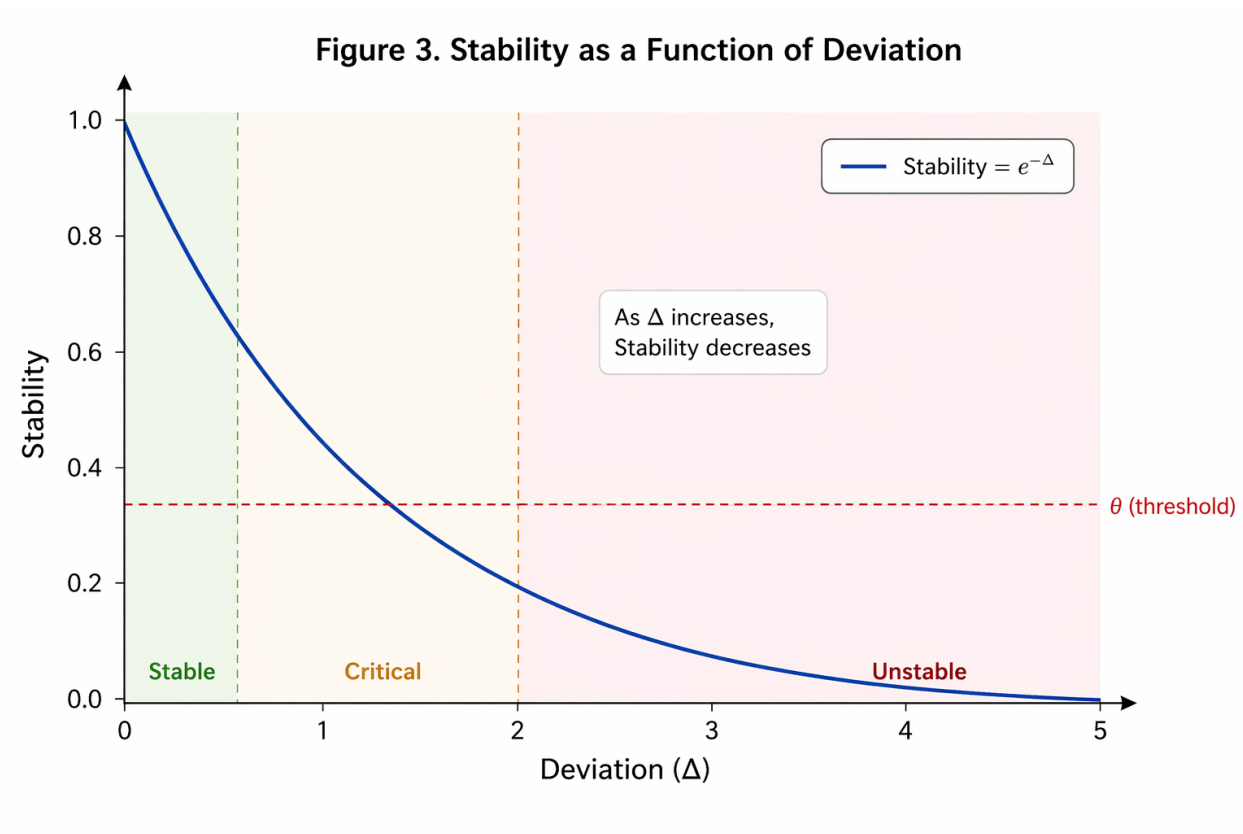

Figure 3: Figure 3. Stability as a Function of Deviation

Deviation:

$$\Delta(S) := \mathbb{E}_{O_1,O_2}\left[d(O_1(S), O_2(S))\right]$$

Stability:

$$\text{Stability}(S) := \exp(-\lambda\Delta(S)), \quad \lambda > 0$$

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5. Collapse into Void

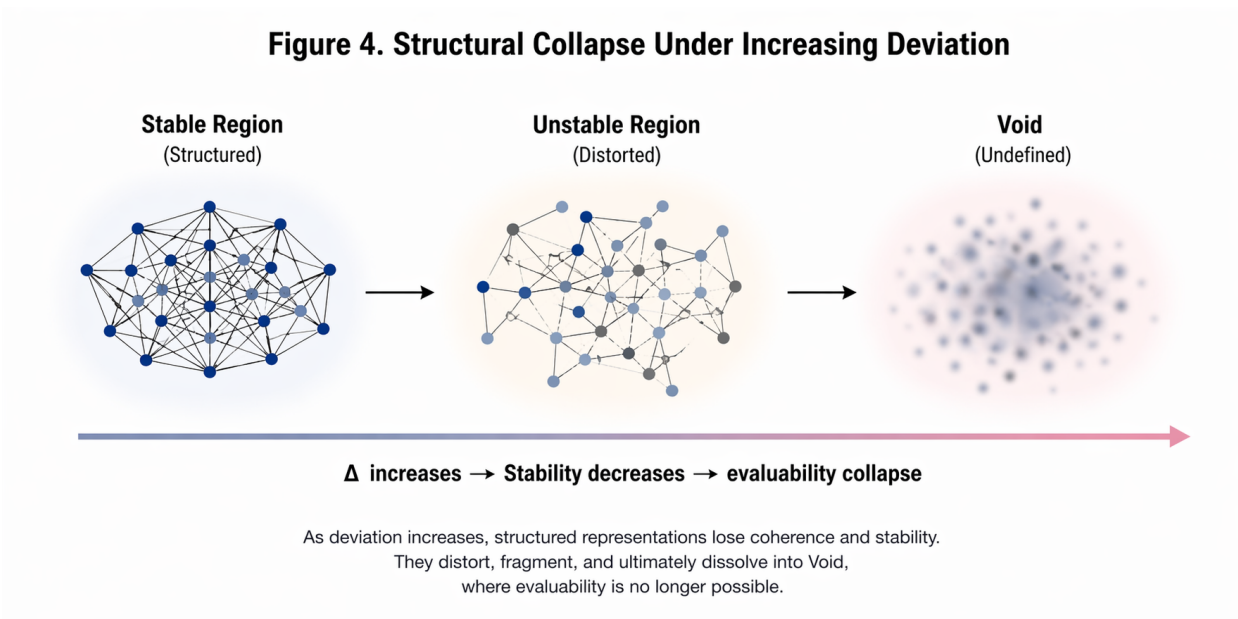

Figure 4: Figure 4. Structural Collapse Under Increasing Deviation

$$\text{Void} := \{X \mid \Delta(X) \text{ undefined or diverges}\}$$

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6. Discontinuous Transition (Jump)

$$\Delta > \theta \Rightarrow O \to O'$$

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7. Identity as Stable Trajectory

$$\gamma : [0, T] \to \mathcal{X}$$

$$\mathcal{J}_\theta := \{\gamma \mid \text{Stability}(\gamma(t)) \geq \theta\}$$

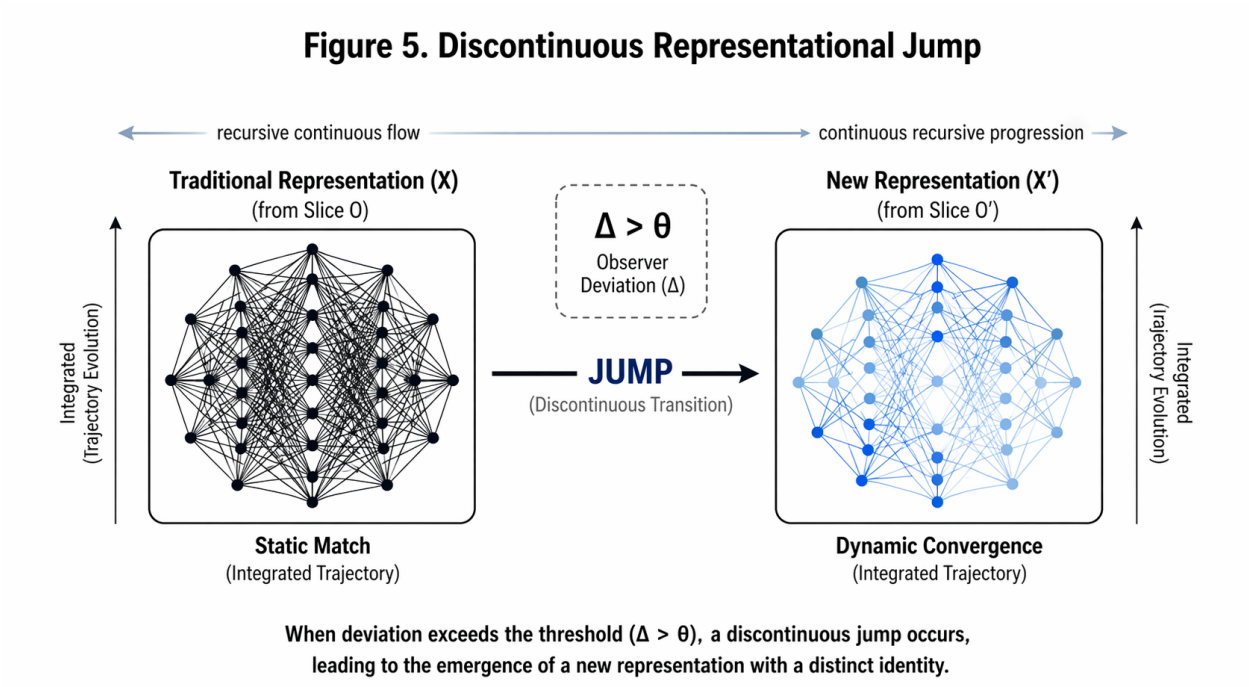

Figure 5: Figure 5. Discontinuous Representational Jump

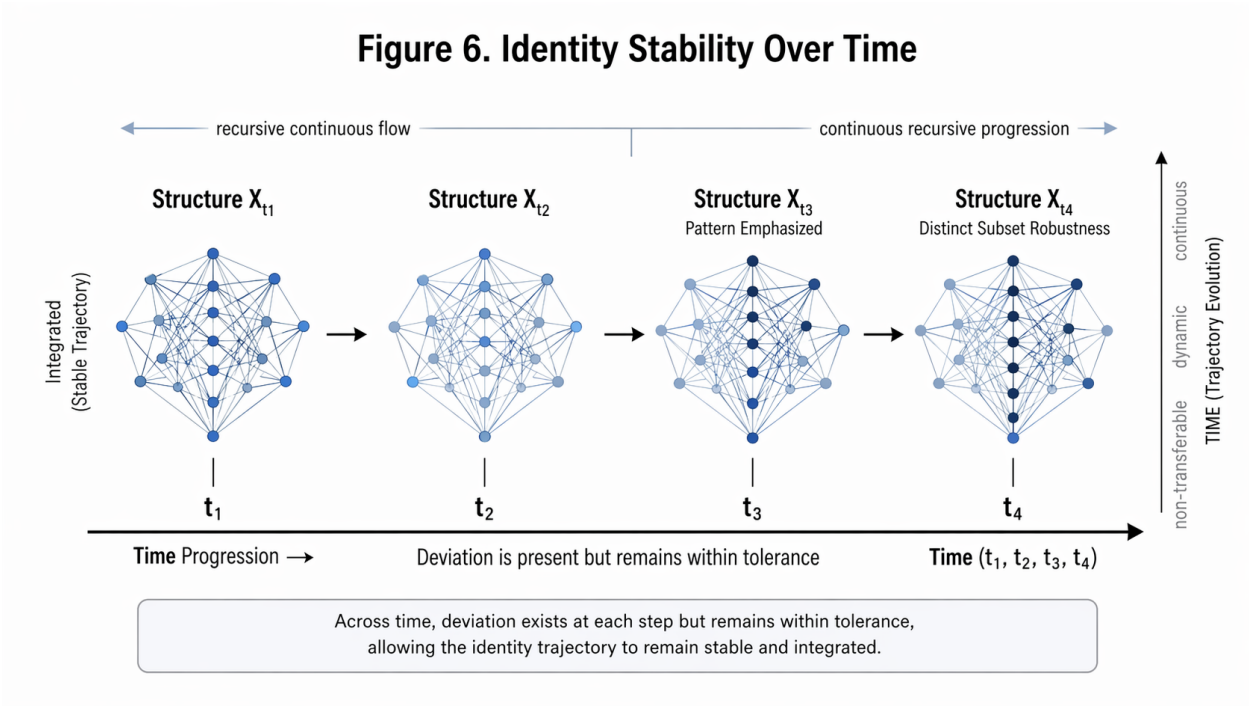

Figure 6: Figure 6. Identity Stability Over Time

8. Conclusion

Gyro Logic provides a unified framework for representation, deviation, and stability.

Appendix A. Formal Definitions

$$O : \mathcal{S} \to \mathcal{X}$$

$$X = O(S)$$

$$\Delta(S) := \mathbb{E}_{O_1, O_2}\left[d(O_1(S), O_2(S))\right]$$

$$\text{Stability}(S) := \exp(-\lambda \Delta(S))$$

$$\mathcal{M}_\theta := \{X \mid \text{Stability}(X) \geq \theta\}$$

$$\text{Void} := \{X \mid \Delta(X) \text{ undefined or diverges}\}$$

$$\mathcal{J}_\theta := \{\gamma \mid \text{Stability}(\gamma(t)) \geq \theta\}$$

Appendix B. Theoretical Results

$$0 < \text{Stability}(S) \leq 1$$

$$\Delta_1 < \Delta_2 \Rightarrow \text{Stability}_1 > \text{Stability}_2$$

$$X \in \mathcal{M}_\theta \iff \text{Stability}(X) \geq \theta$$

$$\Delta(X) \to \infty \Rightarrow \text{Stability}(X) \to 0$$